

Luke Conibear

laconibear@gmail.com | lukeconibear.com | [LinkedIn](#) | UK

Overview

Generalist engineer specialising in simplifying complex scientific software systems across software engineering, AI, cloud infrastructure, and atmospheric science.

Experience

Senior Software Engineer, Tomorrow.io, US company - UK (Remote) | 2022–present

- Primary technical contributor to a department-wide cloud-efficiency initiative that reduced spend by 25% in 2025, through scientific-data compression, capacity optimisation, scale-to-zero, rightsizing, data transfer reduction, and revised data retention policies.
- Identified recurring Severity-1 failure modes and led preventive engineering work associated with a 35% reduction in MTTR from 2024 to 2025.
- Designed, deployed, and maintained production microservices and data pipelines in Python, Rust, and Go for ML and physics-based forecasting workflows, operating within a highly asynchronous distributed team ([technical talk](#)).
- Led the design and delivery of MLOps infrastructure spanning Azure ML, AWS SageMaker, and GCP Vertex AI.
- Led simplification and modernisation of legacy forecasting applications, improving reliability, maintainability, and operational resilience.
- Led department-wide adoption of AI-assisted engineering practices including development, code review, skills, automations, and a knowledge agent.
- Maintain an event-driven distributed system (Go) coordinating multi-cloud weather-data pipelines across AWS, Azure, and GCP, using PostgreSQL-backed dependency tracking.

Research Software Engineer (Machine Learning), University of Leeds, UK (Remote) | 2021–2022

- Developed and taught course ([recording](#)) on machine-learning emulators for numerical atmospheric models.
- Authored and maintained an open-source course for [high-performance Python](#) and [machine learning](#).

Research Fellow, University of Leeds, UK (Remote) | 2018–2021

- Engineered [AI emulators](#) to act as fast proxies for complex atmospheric simulations, distributing training across high-performance compute clusters.
- Maintained and supported [WRFotron](#), an open-source automation tool for atmospheric models, managing GitHub releases, [documentation](#), and a community of users.

Education

- **PhD**, Ambient Air Quality and Human Health, University of Leeds | 2015–2018
- **MSc**, Bioenergy, University of Leeds | 2014–2015
- **BEng**, Mechanical Engineering, University College London | 2007–2010

Publications

- Lead-author of [9 academic papers](#) (h-index of 21 with 1,600+ citations).

Tools

- **AI-assisted development:** Cursor, Codex.
- **Languages:** Python (scientific computing and ML), Rust, Go, Bash.
- **Cloud:** Azure (Azure ML, AKS, Blob Storage, Event Grid); GCP (Vertex AI, GKE, GCS, Pub/Sub); AWS (SageMaker, S3).
- **Platform and operations:** GPUs, Kubernetes, Docker, Terraform, Argo CD, GitHub Actions, KEDA, Datadog, PagerDuty, Git (GitHub).